TOM LACLAVÈRE

Doctor in Astrophysics · Software Engineer – Numerical Simulation & HPC

Doctor in astrophysics and software engineer, specialized in numerical simulation and performance optimization. Lead developer of simulation and data reconstruction pipelines for the QUBIC experiment, deployed on HPC clusters. Seeking a position in numerical simulation or HPC.

PROFIL

- tomlaclavere@gmail.com
- 0643382043
- Permis B/A2
- TomLaclavere
- Website
- LinkedIn

SKILLS

- Languages: C++ (advanced), Python (advanced), Fortran (intermediate)
- HPC: OpenMP, TBB, MPI (mpi4py), SLURM
  ↳ familiar with: SYCL, Thrust, Kokkos, CUDA
- Performance: Profiling (perf, Malt, Maqao), Optimisation, Vectorisation
- Tools: Git, CMake, Docker, Linux, CI/CD, unit testing.
- Data Analysis: Scipy, Pandas, sklearn, PyTorch, JAX

SOFT SKILLS

- Rigor and work organisation
- Autonomy and adaptability
- Written and oral communication
- Ability to explain complex technical concepts
- Technical ownership and sense of responsibility

LANGUAGES

- French (native)
- English (professional)
- Japanese (beginner)

HOBBIES

- Sports: Boxing, badminton
- Programming, video games, motorcycling, hiking

EXPERIENCE

2023-2026
CNRS | Université Paris-Cité | APC Laboratory

PhD - QUBIC Experiment

- Lead developer of the simulation, reconstruction and data analysis pipeline (qubicsoft, > 30k LoC, > 500 commits).
- Large-scale linear system solving: numerical methods & machine learning.
- Parallelisation (mpi4py, OpenMP) and SLURM deployment (up to 20 nodes, 100-200 GB/run).
- Development of calibration and noise analysis tools for data validation.

2025-
GitHub Project

HPC / Simulation Project - 3DPhysicsEngine (C++23)

- 3D rigid body simulation engine: custom math library, two-phase collision pipeline, three integrators (Euler, Verlet, RK4)
- Benchmarks: 150+ configurations (solver × time step), convergence, performance and energy drift quantified
- Current phase: CPU optimization (SIMD, OpenMP); roadmap: CUDA/SYCL, MPI

EDUCATION

2023-2026

PhD in Fundamental Physics - Astrophysique

- QUBIC Data Analysis: Realistic astrophysical components reconstruction and atmospheric mitigation using spectral imaging (APC)
- Goal: Constrain Inflation parameters by detecting primordial CMB B-modes, separating astrophysical foregrounds and atmospheric contamination via spectral imaging.

2025

Gray Scott Summer School - HPC (CC-FR/LAPP)

- TBB, OpenMP, OpenMPI, SYCL, CUDA, Thrust
- C++, Fortran, Rust, Python, Julia

2023

Master Nuclei, Particles, Astroparticles & Cosmology (NPAC) - Mention bien (Université Paris-Cité)

2021

Bachelor's in Fundamental Physic - Mention Très bien (Université Paris-Cité)

2018-2020

CPGE, Physics-Chemistry - Admitted to Centrale (Cité Scolaire Bertran-de-Born)

TEACHING

2024 - 2026

Adjunct Teaching Assistant

CentraleSupélec
- Cosmology - 1ère année - Tutorials (35h) - 2026

Université Paris-Cité
- Physique Numérique - Master 1 - Tutorials (36h) - 2024 & 2025

Ecole d'Ingénieur Denis Diderot
- Champs Electromagnétique - 1ère année - Tutorials (24h) - 2024
- Bruits - 2ème année - Lab sessions (24h) - 2024

PUBLICATIONS

- Atmosphere mitigation in Time-Ordered-Data using Spectral Imaging with QUBIC instrument : ongoing
- Spectral Imaging with QUBIC: building frequency maps from Time-Ordered-Data using Bolometric Interferometry (arXiv:2409.18698)
- Spectral Imaging with QUBIC: building astrophysical components from Time-Ordered-Data using Bolometric Interferometry (arXiv:2409.18714)
- Neural-Network Map-Making : ongoing
- Contribution to the 2024 Cosmology session of the 58th Rencontres de Moriond: Bolometric interferometry and spectral: a QUBIC overview (arXiv:2406.15414)
- Contribution to the 2026 Cosmology session of the 59th Rencontres de Moriond: ongoing

CONFERENCES & TRAININGS

- 59th Rencontres de Moriond, Cosmology session, 2026, Italie, La Thuile
  - Presenter, "Spectral Imaging with QUBIC: Component separation methods using Bolometric Interferometry and application on atmospheric mitigation". (20 min)
- Gray Scott School, 2025, France, Annecy
  - Summer School, "The GRAY SCOTT SCHOOL 2025 - Revolutions will be a deep dive into High Performance Computing, computing optimisation, profiling, and software engineering, to guide you through important topics such as CPU/GPU architectures, Unit Tests, Computing Precision, Memory Allocation and profiling, with modern C++, Rust, Fortran and Python languages, and libraries such as Sycl, EVE, Vulkan, CUDA, Thrust, PyTorch."
- Machine learning in Python with scikit-learn, 2025, Inria, online
  - Training session, Data Analysis with scikit-learn
- Groupement De Recherche in Cosmological Physics (GDR CoPhy) Episode 3, 2025, France, Paris, ENS
  - Presenter, "Spectral Imaging with QUBIC: Component separation methods using Bolometric Interferometry". (25 min)
- Scientific Python Training CC-IN2P3, 2025, France, Lyon
  - Training session, Python training focused on High Performance Computing with Python and utilisation of IN2P3 Computational Center (using SLURM)
- Ecole Rodolphe Cledassou, 2024, France, Hendaye
  - Summer School, "L'école Rodolphe Cledassou forme les jeunes chercheurs francophones à la science des futurs grands relevés cosmologiques. Son fonctionnement est assuré par les organismes et instituts français impliqués dans la mission spatiale Euclid".
- 58th Rencontres de Moriond, Cosmology session, 2024, Italie, La Thuile
  - Poster, "Bolometric interferometry and spectral: a QUBIC overview"